

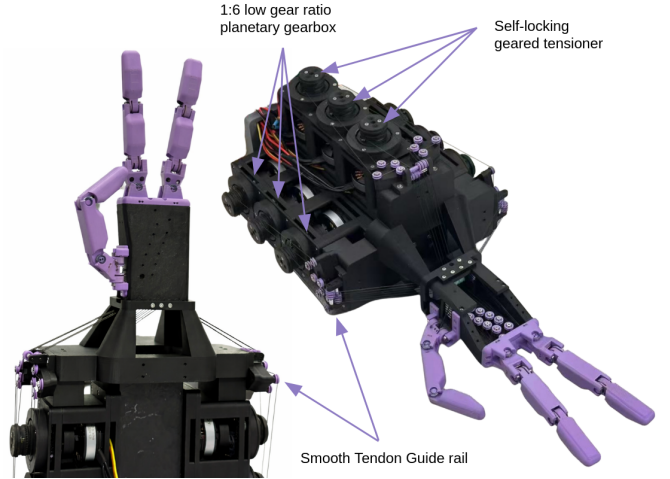
PropHand: A Force-Aware Open-Source Dexterous Robot Hand

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Abstract—Dexterous manipulation is an emerging capability for general-purpose robots, but scaling learning-based policies to contact-rich tasks is bottlenecked by hardware that is either expensive to instrument or poorly suited for force-aware control. In particular, cost-effective force sensing is critical for robot policy learning, yet most open-source dexterous hands lack reliable joint-torque sensing and torque control due to underactuation, nonlinear and configuration-dependent transmissions, high gear ratios that obscure backdrivability, and limitations in commodity electronics and firmware. This work presents PropHand, an open-source tendon-driven dexterous hand designed to make force-aware manipulation accessible and reproducible. PropHand combines mechanically transparent tendon routing with a near-linear actuator-to-joint mapping, enabling consistent calibration and low-bias force inference from motor current. A low gear ratio and torque-controlled actuation improve backdrivability and torque transparency, allowing high-frequency proprioceptive joint-torque sensing without specialized tactile skins. In experiments, PropHand accurately and repeatably estimates force from motor current (from 150 to 350 g with an average error of 6.3%), regulates contact force through torque control (reaching target forces from 0.5 to 4.0 N with an average error of 2.94%), and successfully performs manipulation tasks requiring dynamic and compliant behaviors.

I. INTRODUCTION

Reproducing human-level dexterity requires not only high-degree-of-freedom kinematics, but also precise regulation of contact forces across fast transitions between free-space motion, initial contact, sliding, and force closure. Recent progress in deep reinforcement learning and imitation learning has enabled impressive vision-based manipulation results [1]–[3], yet contact-rich tasks remain fragile under occlusion, high-speed interactions, and small modeling errors [4]–[6]. Crucially, many everyday and industrial behaviors require regulating the magnitude and direction of interaction forces while the nominal pose changes only subtly—e.g., scraping or wiping residue on surfaces, polishing or cleaning where a target normal force must be maintained, twisting a light bulb, or assembling snap-fit parts where success hinges on force during constrained motion. In these regimes, vision-only perception and vision-only demonstrations are often insufficient because contact state and force evolution are weakly observable in videos and images; recent work on torque and force-aware learning explicitly shows that incorporating torque and force feedback and compliant force-executing control substantially improves performance in contact-rich manipulation compared to vision-only baselines, bringing up the need for high quality and accessible force-involved perception for robot learning infrastructures. [7]–[12].



(a) PropHand hardware overview.



(b) Nut turning (c) Tool interaction (d) Everyday grasp

Fig. 1: PropHand: a low-cost open-source dexterous robot hand for force-aware manipulation. PropHand enables accurate contact-force estimation and control using proprioceptive motor-current sensing. A low gear ratio and low-friction tendon-driven transmission improve torque transparency and backdrivability, enabling reliable force interaction without dedicated tactile sensors. (a) Overview of the PropHand hardware platform. (b–d) Representative manipulation behaviors enabled by the system, including nut turning, tool interaction, and dexterous everyday grasping.

Despite this need, the community faces a persistent hardware and data bottleneck. Most force-aware learning results are still arm-centric [11], [12], in part because existing dexterous hands rarely provides transparent, accurate, and high frequency joint torque control and sensing feasible enough to generate meaningful data for policy training. Many open-source dexterous hands prioritize robustness and ease-of-assembly but provide limited force observability and control due to underactuation, friction, and non-backdrivable transmissions [13], [14]. Many commercial hands offer precise

motion with tactile force sensing but at high cost, making them inaccessible to many labs and individual researchers. Additionally, when commercial parts fail, repair can be expensive and require significant downtime. With a low-cost alternative, we aim to make it facilitate the broad deployment of force-aware hands, enabling large-scale diverse data collection needed to learn robust contact-rich policies.

We propose PropHand, a tendon-driven, direct-torque-controlled, highly backdrivable robot hand with high-frequency joint-torque sensing, built from 3D-printed and commercially available components with a total cost of less than 1000 USD (less than 70 USD per DoF including motor, driver board, and encoders). As shown in Fig. 1, PropHand enables accurate force estimation and compliant interaction through torque-transparent actuation and proprioceptive motor-current sensing. In experiments, PropHand accurately estimates external force with an average 6.3% error based on motor current and achieves target force control with as little as 1.53% error. Key contributions include:

- An open-source 3D-printed cable-driven dexterous hand.
- A specialized tendon-driven, low-friction, low-gear-ratio transmission system for dexterous hand applications, optimized for torque-sensing transparency.
- An integrated firmware and software system that enables high-frequency torque sensing and supports torque control as well as impedance-based position control.

II. RELATED WORK

A. Open-Source Robot Hands

Open-source dexterous hands have substantially improved accessibility, but most existing platforms are not designed to provide transparent, high-bandwidth joint-torque sensing and torque control needed for force-aware learning. The LEAP Hand [15] is a widely used, low-cost open platform built around DYNAMIXEL actuators and supports current-based control modes; however, the high gear ratio used in DYNAMIXEL motors (e.g., $\sim 200:1$ reduction) and low communication rate make it infeasible to make meaningful torque estimates from current signals. ORCA [16] is a tendon-driven open-source hand that emphasizes reliability and rapid deployment, and it incorporates integrated tactile sensing; yet its tendon transmission and remote actuation are primarily optimized for dexterous position control and robustness, and the high-friction tendon-driven dynamics as well as tendon forces that arise from joint coupling make accurate per-joint torque inference from actuator-side signals difficult. Furthermore, ORCA has the same current-sensing limitations as LEAP due to the use of DYNAMIXEL servos. RUKA [17] provides an affordable open-source humanoid hand with 15 underactuated DoF and explicitly addresses control by learning joint-to-actuator mappings from motion-capture data; while effective for generating dexterous motion, underactuation and tendon coupling make precise, model-based joint-torque estimation ill-posed without additional sensing and identification. Beyond hand-like platforms, TriFinger [18] provides a well-known open-source system for learning dexterity; however,

it is comparatively large and not intended for fine, anthropomorphic in-hand manipulation. At the same time, its large timing-belt transmission makes it practical to integrate high-performance control and leverage motor-current signals for torque-related feedback in a robust lab setup. At the other end of the design spectrum, ILDA [19] demonstrates an integrated linkage-driven anthropomorphic hand with impressive capability. However, its mechanical design relies on precision-machined components and expensive actuators (e.g., Maxon motors on the order of 300 USD per DoF), which increases fabrication cost and makes replication more difficult for typical research labs. Finally, the RBO Hand 3 platform [20] highlights the advantages of soft dexterous manipulation, but its soft, highly nonlinear deformation and contact dynamics make modeling and learning-based control more challenging compared to articulated rigid-link hands when the goal is accurate, high-bandwidth torque/force inference from proprioception.

These open-source hands motivate PropHand to fill in a gap: a low-cost open-source hand that prioritizes transmission transparency and an integrated control stack for convenient, high-frequency joint-torque sensing and torque/impedance control to support learning with torque and force modalities.

B. Proprioceptive Force Estimation from Joint Torques

Proprioceptive contact-force estimation leverages robot dynamics to infer external interaction forces from joint-torque information (measured by joint-torque sensors or estimated from motor currents). In particular, the torque produced by a motor is proportional to the torque-axis current under field-oriented control, which can be written as

$$\tau_m = K_t i_q, \quad (1)$$

where τ_m is the motor torque, K_t is the motor torque constant, and i_q is the torque-axis current.

External contact wrenches can be related to joint torques through the Jacobian transpose relationship,

$$\tau_{\text{ext}} \approx J(q)^\top f_{\text{ext}}, \quad (2)$$

where $J(q)$ is the Jacobian matrix of the finger and f_{ext} denotes the external contact force expressed in task space. Van Damme, et al. [21] established that meaningful end-effector force estimates can be recovered even when actuator-side torque measurements are noisy, by combining model-based prediction and appropriate filtering. More recent systems further demonstrate that motor-current signals can be exploited to estimate Cartesian contact forces/torques and close the loop for force control without a dedicated wrist force/torque sensor [22]. Complementary model-based estimators improve robustness under modeling errors by combining parametric structure with nonparametric components and disturbance filtering, e.g., semiparametric models with disturbance Kalman filters [23]. Some approaches augment proprioception with additional sensing modalities, such as using cameras together with electrical current to estimate interaction forces, highlighting a trade-off between sensing complexity and estimator robustness [24]. Recent learning-and-control systems explicitly

adopt these principles for practical manipulation: FILIC [12] introduces a cost-effective end-effector force estimator that uses joint-torque measurements together with model-predicted torques (from a digital twin) to estimate contact force when a wrist force/torque sensor is unavailable, and similar sensorless force estimation has been demonstrated in high-stakes domains such as robot-assisted surgery on the da Vinci research kit [25].

A key challenge is noise and model mismatch: inverse dynamics requires accurate inertial/friction terms, and naive differentiation of velocity to obtain acceleration can amplify sensor noise. To mitigate this, generalized momentum observer (GMO) and related momentum-residual methods estimate external torques using momentum dynamics and filtering, improving robustness without relying on direct acceleration measurement [26]. However, for conventional robot arms with long and massive links, acceleration-dependent inertial terms can still dominate the residual during fast motion, degrading force estimates in dynamic manipulation. This motivates our design choice: we use a tendon-driven hand and explicitly minimize linkage mass so that inertial effects (and therefore acceleration-induced estimation error) are reduced, improving the signal-to-noise ratio of torque-based contact-force inference during high-frequency interaction.

III. DESIGN AND METHOD

A. Design Principles

We design PropHand to support force-aware manipulation by prioritizing three constraints.

First, we aim at high transmission force transparency. The tendon routing and drivetrain are designed to minimize friction, hysteresis, and configuration-dependent bias so that motor-current-derived torque can be mapped to joint torque and external force with low distortion.

Second, we emphasize low cost and open-source reproducibility. We thus limit the design to using 3D-printed structural parts and commercially available components, with all key parts purchasable from reliable vendors (e.g., T-Motor for motors and commodity suppliers such as Amazon for bearings and hardware). This reduces fabrication burden and enables reproducible assembly.

Third, we target high-frequency proprioceptive force sensing to support differentiable, dynamics-based inference of external interaction. By measuring torque-axis current at high rate and leveraging robot’s dynamic model, external force can be estimated, providing a frequency force modality for learning.

Table I situates PropHand relative to representative dexterous hands in terms of gear ratio and force transparency.

B. Torque-Transparent Tendon Redirecting

To maximize torque transparency and reduce configuration-dependent bias in tendon length and moment arm, we route tendons through the joint centers of rotation wherever possible [16]. Specifically, we design the tendon path to pass through the center of rotation for both DoFs of the MCP joint and for the single DoF of the PIP joint, so that joint angle and torque

TABLE I: Comparison of dexterous hands with explicitly stated gear ratios, adapted from Origami Robotics [27]–[32]. We compare Gear Ratio, where lower ratio improves back-drivability, Force Transparency, which improves feedback control, and availability as Open-Source (OS).

Hand	DoF	Gear Ratio	Force Transp.	OS
Allegro Hand	16	369:1	Low (blind)	No
TESOLLO	20	100:1	Low-Medium	No
SharpaWave	22	200:1	Low	No
Wuji Hand	20	100:1	Low-Medium	No
Psyonic	6	100:1	Low-Medium	No
Origami Hand	21	15:1	High (motor current)	No
LEAP Hand	16	288:1	Low (blind)	Yes
RUKA Hand	15	288:1	Low	Yes
ORCA Hand	17	288:1	Low-Medium	Yes
PropHand (Ours)	9	6:1	High	Yes



Fig. 2: Tendon pathway design for torque-transparent routing. Tendons are routed through the centers of rotation of the MCP (two DoFs) and PIP joints to reduce configuration-dependent moment-arm variation and minimize inter-joint coupling.

coupling due to rotation in adjacent joints can be eliminated (Fig. 3).

Furthermore, we apply multiple design details to minimize friction alongside the tendon pathways. Each redirection inside fingers uses polished dowel pins with lubrication. For tight turns where sliding friction cannot be reduced to a feasible level, we introduce a bearing-based guiderail assembled from off-the-shelf miniature bearings, converting sliding contact into rolling contact along the tendon pathway (Fig. 2).

Fig. 3 illustrates the tendon routing pathway for one finger; the guide rails are designed to prevent tendon-tendon and tendon-structure contact, thereby reducing sliding interference and minimizing friction.

C. Low Gear Ratio 3D-Printed Planetary Gearbox

We adopt a low-ratio planetary gearbox to achieve sufficient torque output and rigidity for the fingers while maintaining torque transparency. Compared to simple spur reductions, planetary gear trains distribute load across multiple planet gears, reducing tooth stress concentration and improving longevity, which is particularly important for 3D-printed gears that are sensitive to localized wear and deformation.

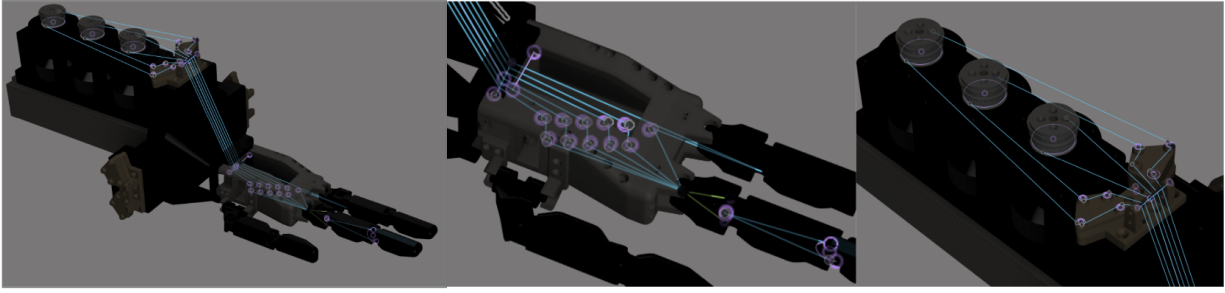


Fig. 3: Tendon pathway configuration rendered in CAD. Tendon guiderails on the wrist flange as well as in the palm separates the tendon from each other and from other hand components.

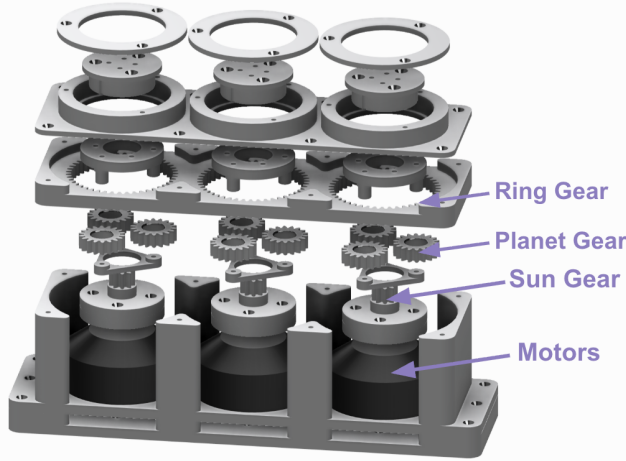


Fig. 4: Rendering of the low-ratio 3D-printed planetary gearbox used in PropHand.

To minimize internal losses and preserve backdrivability, we use press-fitted ball bearings throughout the drivetrain: the carrier is supported on bearings, and each planet gear rotates on its own bearing, converting internal sliding contact to rolling contact and thereby reducing friction and hysteresis inside the gearbox. To account for dimensional errors, strain, and surface roughness introduced by fused-filament fabrication, we generate the gear profiles with a designed backlash of 0.15 mm, which empirically reduces binding while maintaining acceptable positional repeatability. Fig. 4 shows a rendering of the proposed planetary gearbox assembly.

D. Electronics and Firmware

PropHand uses an AS5048A magnetic encoder for joint position sensing and an STM32G431-based B-G431B-ESC1 motor driver board for actuation and low-level control. Compared to other controllers like the ODrive series controllers [33], the B-G431B-ESC1 is lower cost (only 18 USD per unit and can be purchased through common suppliers like DigiKey and Mouser). In addition, the B-G431B-ESC1 is designed for motor-control applications and supports shunt-based current sensing for Field-Oriented Control (FOC). The resulting current-sense signals are sampled by the STM32G431’s on-

chip ADCs (two fast 12-bit ADCs), enabling reliable phase-current measurement and high-rate torque-axis current (i_q) estimation [34]. For communication, the board exposes multiple control/monitoring interfaces (e.g., UART and CAN) [34]; since the STM32G431 includes an on-chip FDCAN controller, the same interface can also be used in CAN-FD-capable deployments when needed [34].

We run custom FOC firmware (included in the open-source release) on the STM32G431 that provides high-rate torque-axis current (i_q) sensing, which serves as the primary proprioceptive signal for torque estimation and torque- or impedance-based position control. The inner FOC loop runs at 4 kHz, and the motor driver communicates directly with a host computer over USB, enabling a 200–500 Hz high-level control/update rate.

E. Proprioceptive Force Estimation from Torque-Axis Current

We estimate external interaction forces from motor torque-axis current by leveraging a simplified inverse-dynamics model. For a finger with generalized coordinates q , the joint-space dynamics can be written as

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) + \tau_f(\dot{q}) + \tau_{\text{ext}}, \quad (3)$$

where τ is the commanded/realized joint torque, M is the inertia matrix, C captures Coriolis/centrifugal effects, g is gravity, τ_f is friction, and τ_{ext} denotes torques induced by external contact.

PropHand fingers are lightweight and many contact interactions of interest involve quasi-static force exertion with low velocity; thus we assume $\dot{q} \approx 0$ and $\ddot{q} \approx 0$ during force regulation, which removes the inertial and Coriolis terms. Furthermore, because the finger links are low-mass and the contact experiments are performed in near-horizontal configurations, the gravity term is small relative to contact torques and is neglected in the quasi-static estimator. Under these assumptions, the measured motor torque can be directly attributed to external interaction and friction:

$$\tau_{\text{ext}} \approx \tau - \tau_f(0). \quad (4)$$

We obtain motor torque from torque-axis current as $\tau_m = K_t i_q$ and map it to joint torque through the transmission ratio and calibrated efficiency model, yielding an estimate $\hat{\tau}_{\text{ext}}$.

Finally, we reconstruct the end-effector wrench (or contact force) using the Jacobian transpose relationship:

$$\hat{\tau}_{\text{ext}} \approx J(q)^\top \hat{F}, \quad (5)$$

where $J(q)$ is the finger Jacobian and $\hat{F}$ is the estimated end-effector force expressed in task space. When the finger is not fully actuated or $J^\top$ is not square, we compute $\hat{F}$ using a (damped) least-squares solution to (5).

IV. EXPERIMENTS

We evaluate PropHand from three perspectives: (i) accuracy of contact force estimation from motor-current-derived joint torques, (ii) closed-loop force tracking performance under force control, and (iii) demonstrations on diverse contact-rich tasks.

A. Force Estimation Accuracy

We evaluate how accurately PropHand can estimate external forces using only torque-axis motor current. Our estimator is based on the quasi-static assumption that $\dot{q} \approx 0$ and $\ddot{q} \approx 0$ during static force exertion; thus inertial and Coriolis terms in (3) are negligible and the measured joint torque is dominated by external contact torque. With a known moment arm R from the joint center to the force application point, we approximate

$$\tau_{\text{ext}} \approx F_{\text{ext}} R. \quad (6)$$

Motor torque is obtained from torque-axis current as $\tau_m = K_t i_q$, and mapped to joint torque through the transmission ratio G (gear ratio), giving

$$F_{\text{ext}} R \approx G K_t i_q, \Rightarrow F_{\text{ext}} \approx \frac{G K_t}{R} i_q. \quad (7)$$

Eq. (7) implies a theoretically linear relationship between measured force and motor current when the transmission is torque-transparent and friction/hysteresis are small. In practice, we directly calibrate the proportionality constant

$$F_{\text{ext}} \approx K i_q, \quad (8)$$

where K absorbs $\frac{G K_t}{R}$ as well as residual losses; after calibration, force is predicted using (8) with no sensing beyond torque-axis current.

We obtain the calibration constant K by collecting paired measurements of torque-axis current and ground-truth contact force using an external force/torque sensor. Specifically, we command the finger to repeatedly press into and lift off the sensor, recording synchronized samples $\{(i_q^{(k)}, F_{\text{ext}}^{(k)})\}_{k=1}^{200}$ spanning the operating force range. We then fit the linear model in (8) via least-squares regression,

$$K = \arg \min_K \sum_{k=1}^{200} \left(F_{\text{ext}}^{(k)} - K i_q^{(k)} \right)^2, \quad (9)$$

and use the resulting K for subsequent contact-force estimation using only torque-axis current, without any additional force sensing.

Leveraging our obtained K , we perform the following tests to quantify estimation accuracy.

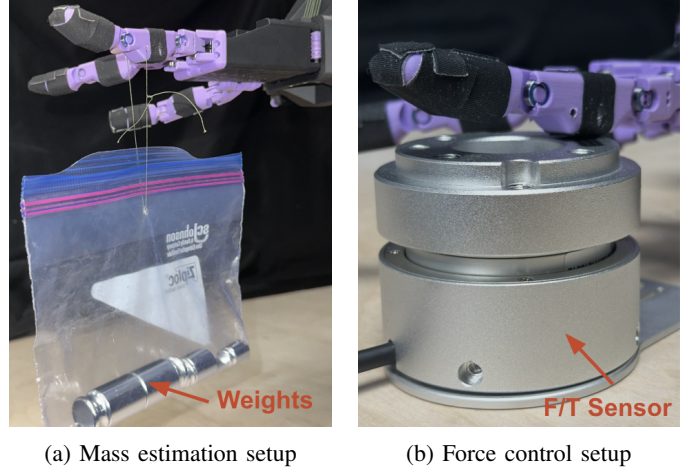


Fig. 5: Experimental setups used in our evaluation. (a) Random-mass generalization test where the finger presses against a scale to measure ground-truth weight. (b) Force control evaluation where the finger regulates contact force against a force/torque sensor.

a) Test 1: Random-mass generalization: To evaluate how well the calibrated model generalizes to unseen loads, we test the system using a set of calibrated weights with known masses. As shown in Fig. 5 (a), the weights are directly suspended from the fingertip so that the gravitational force acts on the finger.

We use a set of standard weights ranging from 150 g to 350 g (in increments of 25 g). For each weight, the finger supports the load at a fixed configuration, and the steady-state torque-axis current is recorded once the system reaches equilibrium.

Using the calibrated mapping in (8), we estimate the corresponding mass and compute the mean percentage error (MPE) across all tested weights:

$$\text{MPE} = \frac{1}{N} \sum_{k=1}^N 100 \cdot \frac{|m^{(k)} - \hat{m}^{(k)}|}{m^{(k)}}. \quad (10)$$

where $N = 9$ denotes the total number of tested mass values. The estimation results for all tested masses are summarized in Table II. Across all mass values, the system achieves a mean percentage error of 6.81%, indicating accurate force estimation across different loads.

b) Test 2: Repeatability: To assess repeatability under identical loading conditions, we select a fixed mass of 275 g and repeat the same measurement multiple times under the same finger pose and contact geometry.

The mean percentage error across trials is computed using the same metric defined in Eq. (10), where $N = 8$ denotes the number of repeated trials.

The repeatability results are reported in Table III. Across repeated trials, the system achieves a mean percentage error of 2.72%, indicating stable and consistent force estimation.

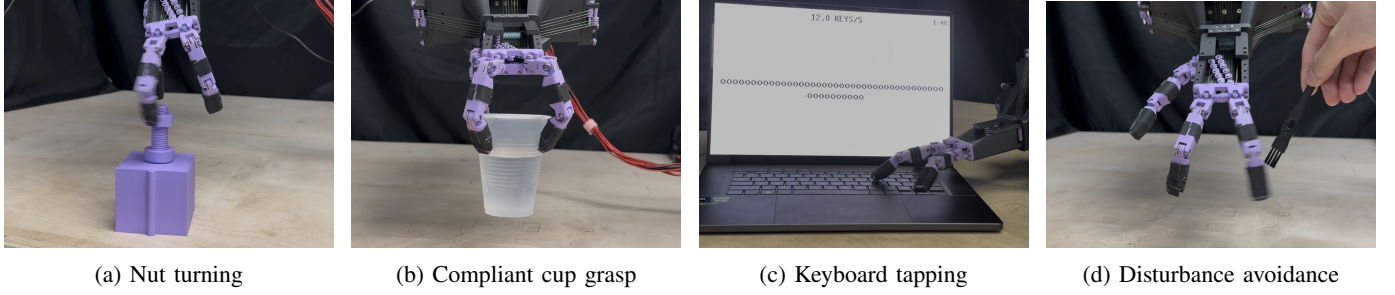


Fig. 6: Representative task demonstrations enabled by high backdrivability and high-frequency torque sensing/control. From left to right: (a) single-finger nut turning, (b) compliant grasping of a cup without deformation, (c) high-speed keyboard tapping, and (d) disturbance-aware reflexive avoidance in response to external forces.

TABLE II: Mass (weight) estimation from torque-axis current for Random-Mass Generalization test. Estimated mass is computed as $\hat{m} = \hat{F}/g$ with $g = 9.81 \text{ m/s}^2$. Percentage error is $\text{PE} = 100 \cdot \frac{|\hat{m} - m|}{m}$.

Scale Mass (g)	Estimation Results	
	Estimated Mass (g)	Percentage Error (%)
150	131.50	12.33
175	138.63	20.78
200	184.51	7.75
225	206.24	8.34
250	250.76	0.31
275	264.68	3.75
300	278.06	7.31
325	324.16	0.26
350	351.68	0.48
Mean Percentage Error (%)		6.81

TABLE III: Mass estimation repeatability test for a fixed load of 275 g. Estimated mass is computed as $\hat{m} = \hat{F}/g$ with $g = 9.81 \text{ m/s}^2$. Percentage error is $\text{PE} = 100 \cdot \frac{|\hat{m} - m|}{m}$.

Scale Mass (g)	Estimation Results	
	Estimated Mass (g)	Percentage Error (%)
275	281.20	2.25
275	266.48	3.10
275	285.37	3.77
275	269.92	1.85
275	283.64	3.14
275	271.15	1.40
275	286.02	4.01
275	268.77	2.26
Mean Percentage Error (%)		2.72

B. Force Control Accuracy

The purpose of this experiment is to evaluate how accurately PropHand can realize a desired contact force at convergence. As shown in Fig. 5 (b), the finger presses against an external force/torque sensor that provides actual contact force measurements. Importantly, the closed-loop force controller uses only motor-current-based proprioceptive perception for feedback; the external force/torque sensor is not used for control and serves only as an evaluation instrument. For each trial, we

record the contact force measured by the force/torque sensor after the interaction has converged. As shown in Fig. 7, the measured force becomes approximately constant near the end of the trial, indicating convergence, and this converged value is documented as the actual force tabulated in Table IV.

We implement a simple outer-loop force regulator that adjusts a joint position command based on the estimated contact force from torque-axis current (Sec. III-D), while the embedded firmware executes low-level PD control. The outer loop runs at 50 Hz over USB.

a) Outer-loop force regulation: We regulate fingertip contact force using a simple two-layer controller (Alg. 1). A high-level outer loop runs on the host computer at 50 Hz and updates a joint position command based on the estimated contact force from torque-axis current, while the embedded firmware executes a low-level PD position controller to track the commanded joint angles. This structure avoids requiring a high-bandwidth force controller on the microcontroller and isolates fast motor control (FOC and PD tracking) from the slower USB communication loop.

At each outer-loop iteration, we read the torque-axis current i_q and convert it to a force estimate $\hat{F} = K i_q$ using the calibration in Sec. III-D. The outer loop then performs a sign-based incremental update on the commanded angle of the tested joint j : if $\hat{F}$ is below the desired force F_{des} , the command is increased by a fixed step $\Delta\theta$ to press further into the sensor; otherwise, the command is decreased by $\Delta\theta$ to relieve contact. This repeated push–release adjustment drives the contact force toward the setpoint despite unmodeled friction and transmission losses. After convergence, we compute the steady-state force estimate $\hat{F}_{\text{ss}}$ as the mean of $\hat{F}$ over a fixed hold window, and report tracking accuracy using the percentage error metric defined below.

b) Evaluation Metrics: We evaluate eight desired force values and repeat the force-reaching procedure three times for each force value, as shown in Table IV. For each trial, we report the steady-state percentage error at convergence:

$$\text{PE}_{\text{ctrl}} = 100 \cdot \frac{|F_{\text{des}} - \hat{F}_{\text{ss}}|}{F_{\text{des}}}, \quad (11)$$

Algorithm 1 Outer-Loop Force Regulation Using Proprioceptive Current-Based Force Estimation

Require: Tested joint j , desired force F_{des} , step size $\Delta\theta$, update rate f

Ensure: Converged force estimate $\hat{F}_{\text{ss}}$ and final command θ^{cmd}

```

1: Record initial joint angles  $\theta^0 = [\theta_1^0, \theta_2^0, \theta_3^0]$ 
2:  $\theta^{\text{cmd}} \leftarrow \theta^0$ 
3: while not converged do
4:   Read torque-axis current  $i_q$ 
5:    $\hat{F} \leftarrow K i_q$  ▷ See Sec. IV-D
6:   if  $\hat{F} < F_{\text{des}}$  then
7:      $\theta_j^{\text{cmd}} \leftarrow \theta_j^{\text{cmd}} + \Delta\theta$ 
8:   else
9:      $\theta_j^{\text{cmd}} \leftarrow \theta_j^{\text{cmd}} - \Delta\theta$ 
10:  end if
11:  Send  $\theta^{\text{cmd}}$  to embedded PD controller
12:  Wait  $1/f$  seconds
13: end while
14:  $\hat{F}_{\text{ss}} \leftarrow \text{mean}(\hat{F})$ 

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TABLE IV: Force control results at convergence. Percentage error is computed per trial as $\text{PE}_{\text{ctrl}} = 100 \cdot \frac{|F_{\text{des}} - \hat{F}_{\text{ss}}|}{F_{\text{des}}}$, and then averaged across trials for each setpoint.

Target (N)	Trial 1 (N)	Trial 2 (N)	Trial 3 (N)	Mean Err (%)
0.50	0.517	0.503	0.522	2.80
1.00	0.945	1.101	1.064	7.33
1.50	1.445	1.489	1.506	1.60
2.00	1.946	2.047	1.978	2.05
2.50	2.442	2.453	2.639	3.25
3.00	2.944	3.062	2.980	1.53
3.50	3.440	3.415	3.477	1.60
4.00	3.947	4.217	4.133	3.36
Mean Percentage Error (%)				2.94

where $\hat{F}_{\text{ss}}$ is the mean measured force over a fixed hold window after convergence.

The results show that PropHand can regulate contact force with high accuracy across the tested force range. As summarized in Table IV, the controller achieves a mean percentage error of 2.94% over all evaluated setpoints. In addition to this overall accuracy, the closed-loop response exhibits stable convergence behavior, including at low-force operation. In particular, Fig. 7 shows that even for a target force as small as 0.5 N, the controller drives the interaction force to a steady value after a short transient, indicating reliable convergence and consistent low-force regulation.

C. Task Demonstrations

To illustrate the breadth of behaviors enabled by high back-drivability and high-frequency torque sensing and control, we present qualitative demonstrations on four representative tasks. These tasks highlight the system’s capability for dexterous manipulation, compliant interaction, high-speed motion, and reactive behavior. Specifically, we demonstrate (1) single-

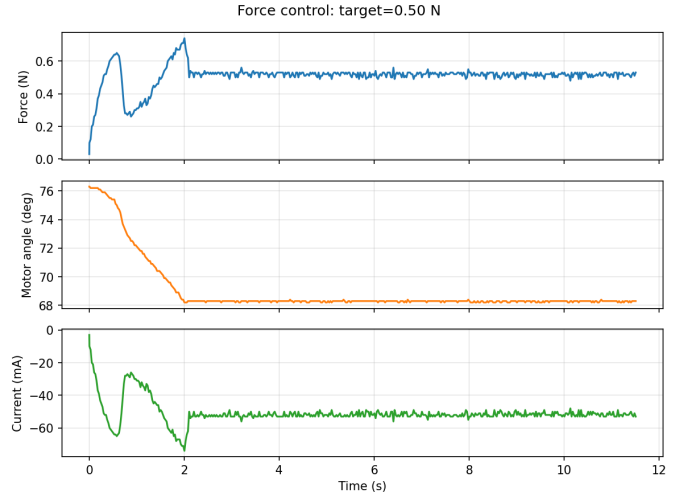


Fig. 7: Convergence of force control to a target contact force of 0.5 N. at $F_{\text{des}} = 0.5$ N. The measured force, motor angle, and motor current are shown from top to bottom.

finger nut turning, where the finger repeatedly applies controlled torque to rotate a nut; (2) compliant grasping of a flexible plastic cup without causing deformation; (3) high-speed keyboard tapping enabled by rapid torque control; and (4) disturbance-aware reflexive avoidance, where the finger detects external forces and actively moves away. Fig. 6 shows representative frames of these demonstrations.

Across these demonstrations, PropHand exhibits strong dynamic force estimation and torque control capabilities. In the single-finger nut-turning task, the finger is able to continuously spin the nut while tracking its gradual downward motion, reaching rotation frequencies up to 8 Hz with minimal loss of contact throughout the process. In the compliant cup-grasping task, the hand successfully grasps a flexible plastic cup without noticeable slipping or visible deformation, indicating that the applied grasp force is both stable and appropriately regulated through force control. In the high-speed keyboard tapping task, PropHand achieves rapid yet precise cyclic motion at frequencies up to 12 Hz, while remaining sufficiently compliant to avoid damaging the keyboard. In the disturbance-aware avoidance task, the system is sensitive enough to perceive extremely light external contact, such as from a soft brush, and react by reflexively moving away. Together, these four tasks validate the hand’s ability to estimate and regulate contact forces in real time, and demonstrate that PropHand is well suited for dynamic and dexterous force-aware manipulation.

V. CONCLUSION

This work presented PropHand, an open-source dexterous hand designed to make force-aware manipulation more accessible through low-cost, reproducible, and proprioceptive hardware. By combining a torque-transparent tendon transmission, low-ratio actuation, and current-based joint torque sensing, PropHand enables external torque estimation from joint torque

current without requiring dense tactile instrumentation. Experiments showed that PropHand achieves competent external torque estimation and accurate torque control performance, supporting its use as a practical hardware platform for force-aware learning in contact-rich manipulation.

Despite these strengths, the current system has several limitations. Because many structural components are 3D printed, long-term rigidity is limited, and fatigue, hysteresis, and wear may increase as the hand is used over time. In addition, while the current proprioceptive approach provides useful information about contact magnitude, contact location is not directly observed. This motivates future integration of cost-effective sensing that can provide richer contact localization, enabling more precise estimation of both contact location and contact magnitude without substantially increasing system cost. Finally, performing more challenging force-sensitive manipulation tasks may require a force-aware teleoperation interface that better matches the hand's sensing and control capabilities, enabling higher-quality demonstrations and improved data collection for learning-based methods.

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